

Consumer Surplus and Measuring Welfare Changes

JEB104 Microeconomics I
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Consumer Surplus and Measuring Welfare Changes

Consumer Surplus
Compensating and Equivalent Variation

Demand for a Discrete Good
Continuous Demand
Why Quasilinear Utility?

Demand for a Discrete Good

Discrete good = only available in integer amounts

Demand can be constructed (staircase):

- If price is too high, there will be no demand at all.
- Below a certain price level (r_1), consumer demands 1 unit of the good.
- Below a different price (r_2), consumer demands 2 units.
- This continues until consumer spends all income on the good.

Reservation Price

The reservation price r_k is the price at which the consumer is just indifferent to consume k or $k - 1$ units of the good.

If $p_2 = 1$, then because of indifference for reservation prices, we can derive the following consumer's utility $u(x, y)$:

- ① $u(0, m) = u(1, m - r_1)$
- ② $u(1, m - r_2) = u(2, m - 2r_2)$
- ⋮
- Ⓚ $u(k - 1, m - (k - 1)r_k) = u(k, m - kr_k)$

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Consumer Surplus and Measuring Welfare Changes

Previously

- We have analyzed price changes a lot so far. If our consumer has a given income to spend and if the price of one good changes, the consumer moves to a different indifference curve. (It is a movement along Walrasian demand.) So changes in prices change welfare (utility) of consumers.
- Last time – using price indices, we could evaluate whether individual is better off or worse off in some cases, but definitely we could not measure precisely the change in consumer's welfare.

Today

- Welfare evaluation of economic changes – we will introduce measures that help us to evaluate changes in consumer's welfare due to changes of prices.
- Indirect utility function – this function expressed the maximum level of utility we could achieve given prices and income, so it could be used to show differences in utilities under different prices. However, utility is measured in utils, so it is hard to imagine what a particular change in utility means.
- Hence, today we will introduce MONETARY measures of welfare changes due to price change.

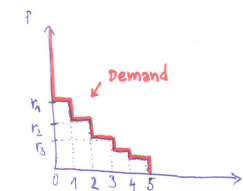
Firstly, we will talk about consumer surplus

- It could be defined as a gain of consumer from buying some good.

Let's start with demand for discrete good.

Discrete good

- Until now we did not (usually) have such assumption, we allowed consumer to spend fractions of units.
- In real life, we often cannot consume fractions of units. Here, it mainly serves for illustration.



Reservation price

- Comes from auction markets
- When someone wanted to buy something in an auction he would be typically aware of a maximum price at which he was willing to buy the good.
- If the best price offered was above this price (reservation price), the buyer declines to buy the good.
- Eventually can be used to describe the minimum price at which someone was just willing to sell some item.

Demand for a Discrete Good: Example

Example: *quasilinear* utility $u(x, y) = f(x) + y$, x discrete good

What are reservation prices?

$$\begin{aligned} u(k-1, m - (k-1)r_k) &= u(k, m - kr_k) \\ f(k-1) + m - (k-1)r_k &= f(k) + m - kr_k \\ r_k &= f(k) - f(k-1) \end{aligned}$$

For which prices is optimal consumption $x^* = 6$?

$$u(x^*) = f(6) + m - 6p \geq f(x) + m - px = u(x)$$

$$\text{If } x = 5: f(6) + m - 6p \geq f(5) + m - 5p \rightarrow f(6) - f(5) = r_6 \geq p$$

$$\text{If } x = 7: f(6) + m - 6p \geq f(7) + m - 7p \rightarrow f(7) - f(6) = r_7 \leq p$$

If 6 units of x demanded, then price p of good x must be: $r_6 \geq p \geq r_7$

Generally:

If n units of x are demanded, then price p of good x must be: $r_n \geq p \geq r_{n+1}$

Constructing Utility from Demand

Reservation prices = differences in utilities = marginal utility of n -th unit

$$r_n = f(n) - f(n-1)$$

Utility of $x = n$ if $f(0) = 0$: Gross benefit/Gross consumer's surplus of x

$$f(n) = \sum_{i=1}^n r_i$$

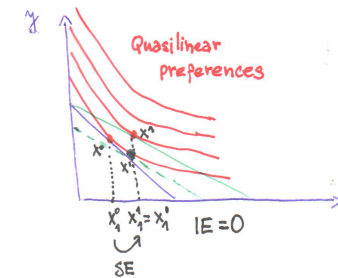
Total utility: $f(n) + m - pn$

Consumer's surplus/Net consumer's surplus: $CS = f(n) - p \cdot n$

- Net benefits from consuming n units of the discrete good x

Quasilinear utility helps

- y are dollars spent on other goods
- Consumption of x does not depend on level of y and m (above certain level of income)
- Interior optimum: $\frac{f'(x)}{1} = \frac{p_x}{p_y}$, income effect is zero
- the picture shows that with decrease of p_1 optimal bundle changes from x^0 to x^1



Budget constraint: $m = p_x \cdot x + p_y \cdot y$, if $p_y = 1$, then $y = m - p_x \cdot x$. This expression for y is plugged in the utility function – if k units of x are consumed and bought for price r_k , then $y = m - kr_k$.

$r_k = f(k) - f(k-1)$, hence the reservation price for k -th unit of good r_k measures exactly the change of utility we derive from consumption of this one extra unit!

For the consumption bundle $x = 6$ to be optimum it must hold that the utility of consuming $x = 6$ must be at least as large as the utility of consuming any other bundle $(x, m - px)$

- we know that Walras law holds, so in the optimum $y = m - 6p$.
- We compare utility of the optimal bundle to utilities of bundles where x is either 5 or 7 and get the range for price of x .

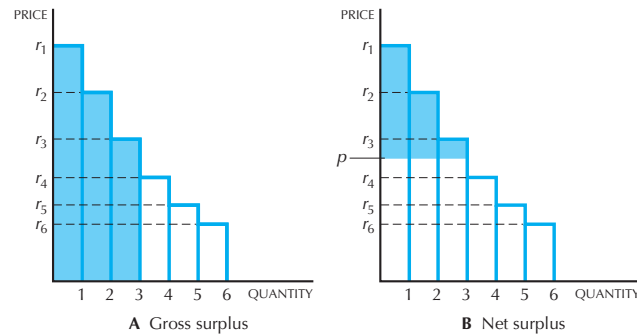
Gross benefit

- Sum of stairs of the figure, sum of reservation prices
- $f(n) - f(n-1) + f(n-1) - f(n-2) + f(n-2) - f(n-3) + \dots + f(1) - f(0) = f(n) - f(0)$

Consumer's surplus (net benefit)

- Gross benefit minus spending on consumption of x

Demand for a Discrete Good



Panel A: Gross surplus $f(3)$

Panel B: Net surplus $f(3) - 3p$

Panel A: Gross surplus $f(3)$

- area underneath the demand curve measures the utility of consumption of that good
- each column represents utility of a particular unit, we can see that marginal utility decreases (each column is getting smaller as we add more units – reservation price we are willing to pay for an additional unit decreases with number of units consumed)

Panel B: Net surplus $f(3) - 3p$

- Deducts how much one pays

Consumer surplus

- We have been considering the case of a single consumer.
- But we can add up each consumer's surplus across all consumers to create an aggregate measure of the consumers' surplus
- CS=convenient measure of the aggregate gains from trade

Consumer Surplus and Continuous Demand

Quasilinear utility function: $u(x, y) = f(x) + y$, y composite good

$$\max_{x, y} f(x) + y$$

$$\text{s.t.} \quad px + y = m$$

Unconstrained maximization: $\max_x f(x) + m - px$

FOC:

$$f'(x) = p$$

Inverse demand function: $p(x) = f'(x) \leftrightarrow \text{price} = \text{marginal utility}$

Area below inverse demand for $x = \text{utility of } x$ (gross surplus of x):

$$\int_0^x p(t) dt = \int_0^x f'(t) dt = f(x) - f(0) = f(x)$$

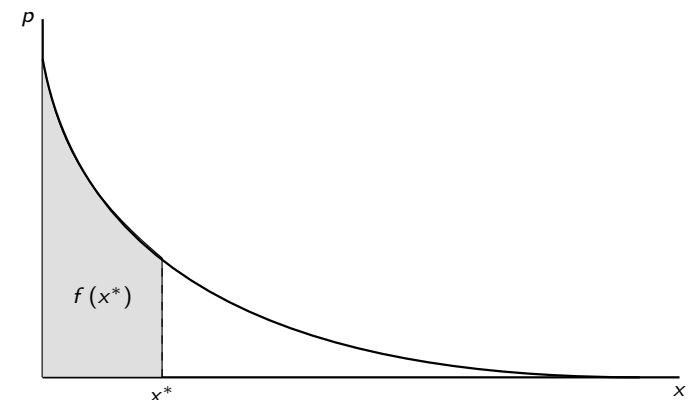
Discrete good case - rather limited, but it is easy to transform the problem for continuous demand

Composite good $\Rightarrow p_y = 1$ (dollars spent on other goods)

We can do unconstrained maximization by plugging the BC into the function

$$f'(x) = MU_x$$

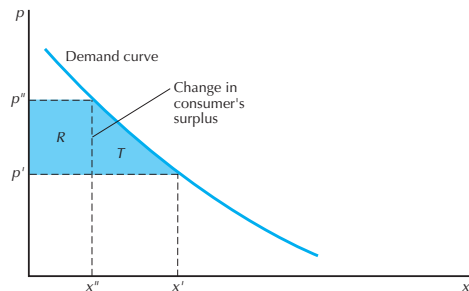
Draw continuous demand:



Inverse demand function shows price as function of x

Change in Consumer's Surplus

Price increase: $p' \rightarrow p''$



Loss of CS by area $R + T$:

- R : consumer still consumes x'' , but each unit is more expensive $x''(p'' - p')$
- T : value of the lost consumption, as the consumer had to decrease consumption from x' to x''

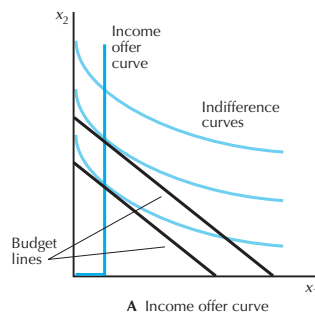
Why Quasilinear Utility?

Special features:

MU_x independent of how much y consumed, MU_y constant

Demand for x independent of $y \Rightarrow$ independent of m : $\frac{\Delta D_x}{\Delta m} = 0$

No income effect = changes in income do not affect demand



Change in consumer surplus: exact measure of utility gain or loss for quasilinear utility function, otherwise only approximation!!

Generally we are more interested in the change in CS than in its level

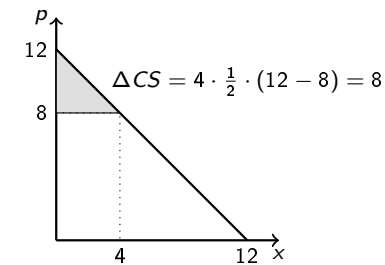
- How does the welfare changes if some policy change inducing price change occurs?

If price increases from p' to p'' , the consumer's surplus will decrease by area $R + T$.

This loss of consumer's surplus measures the loss of utility properly only if preferences are quasilinear!

Example: utility function $u(x, y) = 12x - \frac{1}{2}x^2 + y$, $m = 100$, $p_y = 1$, find optimal x^*

- $MRS = \frac{12-x}{1} = \frac{p_x}{1}$, so inverse demand function is $p_x(x) = 12 - x$ and $x^* = 12 - p_x$
- If $p = 12$, $x^* = 0$ and $u(x, y) = m = 100$.
- If $p = 8$, $x^* = 4$ and $y = m - p_x x^* = 100 - 8 \cdot 4 = 68$, then $u(x, y) = 12 \cdot 4 - 8 + 68 = 108$.
- So change in utility due to change in price is 8.



Quasilinear preferences

- The price you are willing to pay for a good x (equal to a marginal utility of a unit of x you get), is equivalent to what you loose on dollars of other consumption y if you buy this extra unit of x . Utility of A utils you derive from consumption of 1 unit of x is the same as utility you derive from A dollars of other consumption.
- Vertical axis p shows what you are losing in terms of good y when purchasing x , it works perfectly for quasilinear preferences because y are dollars of other consumption and MU_y is constant.

Usually MU of one good depends on how much of other goods we consume

- $u(x, y) = x^a x^b$, $MU_x = a x^{a-1} y^b$
- So MU_x (reservation price) is usually affected by how much money consumer has for consumption of other goods, remember that in the optimum $MU_x/p_x = MU_y/p_y$
- If price of good x decreases, we will not only buy more x and measure change in utility using the area under the demand curve for x , but due to increase in real income, we also have more money for consumption of other goods y and infer additional utility from higher consumption of y (if y is a normal good)
- Area below demand curve cannot be said to measure utility of good x properly.

Monetary Measures of $\Delta u(x)$: Compensating and Equivalent Variation

$\Delta p \Rightarrow \Delta u(x)$ How can we monetarily evaluate this change?

Price decrease ($p^1 < p^0$) $\Rightarrow$ increase in utility

- Which Δm would be welfare equivalent to the price change, given p^0 ?
- How much would we need to give the consumer to reach the utility after the price change, given the old prices?
- How much would consumer be willing to pay to get into situation p^1 ?

Price increase ($p^1 > p^0$) $\Rightarrow$ decrease in utility

- Which Δm would compensate the price change under new prices?
- How much would we need to give the consumer to reach the utility before the price change, given the new prices?
- How much would consumer be willing to receive to get into situation p^1 ?

Compensating Variation (CV)

Compensating Variation

The amount of money we have to give the consumer *after* the price change to make him just as well off as he was *before* the price change.

Bring the consumer to his original utility level under new prices.

Price increase $\rightarrow$ CV positive, price decrease $\rightarrow$ CV negative

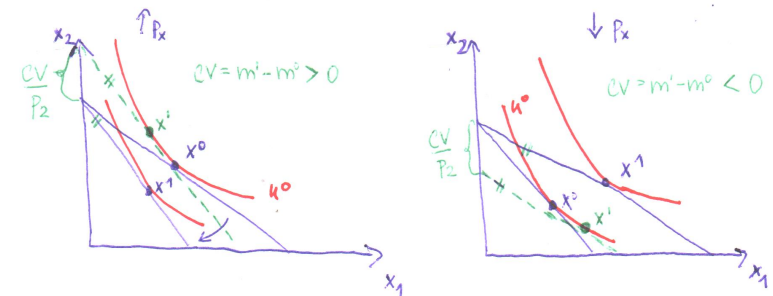
Important note: Different definitions related to sign in different textbooks!!!

When prices change, utility changes as well

- We can imagine a monetary compensation would make the consumer indifferent between situations before and after the price change

Compensating variation

- So we first change the prices, and then we compensate the consumer for this price change.
- We will bring our consumer back to his original indifference curve under new prices.



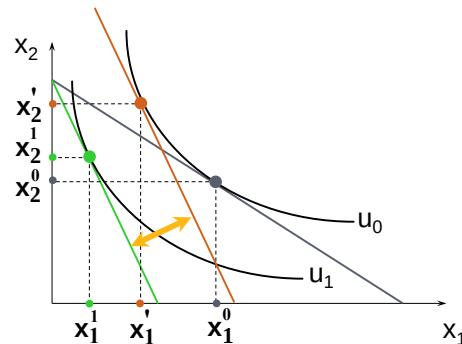
- If price increases, we need to give our consumer extra money ($CV > 0$) to bring him back to his original indifference curve (u^0) to point x' . Under new prices, our consumer has to have income m' (dashed green line) to reach original level of utility. Notice that points x^0 and x' are points on Hicksian demand $h_1(p_1, \bar{p}_2, u^0)$. Difference between the new compensated income m' and the original income m^0 is the Compensating Variation.
- If price decreases, we need to take away some amount ($CV < 0$) to bring him back to his original indifference curve (u^0) to point x' . Now the compensated income m' is lower than the original income m^0 .
- If x_2 is a composite good (i.e. $p_2 = 1$), CV is measurable also directly on vertical axis, otherwise we need to multiply a given distance on vertical axis by p_2 .

Different definitions in different textbook:

- Gravell, Rees, Koubek - the amount of money we have to *take away* from the consumer.
- MasCollé in Advanced Micro - net revenue of a social planner who compensates consumer.

Compensating Variation

Price of x_1 increases: $p_1^0 \rightarrow p_1^1$, p_2 fixed



$$m^0 = p_1^0 x_1^0 + p_2 x_2^0 = p_1^1 x_1^1 + p_2 x_2^1 = e(p_1^0, p_2, u^0) = e(p_1^1, p_2, u^1)$$

$$m' = p_1^1 x_1' + p_2 x_2' = e(p_1^1, p_2, u^0)$$

$$CV = m' - m^0$$

Equivalent Variation (EV)

Equivalent Variation

The amount of money we have to take away from the consumer *before* the price change to leave him as well off as he would be *after* the price change.

Bring the consumer to his new utility level under old prices.

Price increase $\rightarrow$ EV positive, price decrease $\rightarrow$ EV negative

Important note: Different definitions related to sign in different textbooks!!!

Note that there is no substantial difference between this and Hicksian decomposition

- The figure looks the same.

Initial income m^0 is:

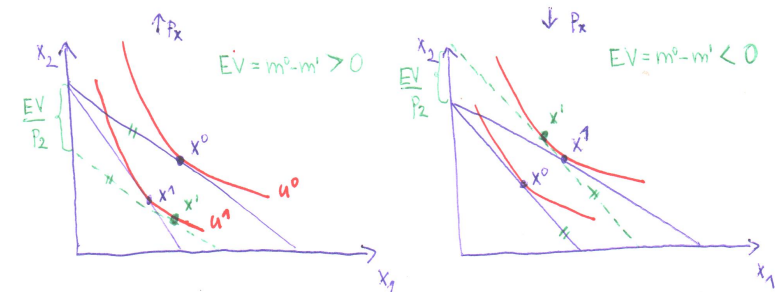
- minimum spending we need to have to reach utility level u^0 under prices (p_1^0, p_2)
- minimum spending we need to have to reach utility level u^1 under prices (p_1^1, p_2)

Compensated income m' is:

- minimum spending we need to have to reach utility level u^0 under prices (p_1^1, p_2)

Equivalent variation

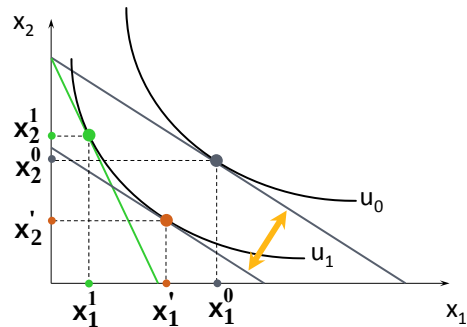
- What income change is equivalent to a price change from the welfare point of view (sign is however opposite – if we take away money, we decrease income, but EV is positive!).
- Before the change, I ask consumer “I will double the price or take your income. How much do I have to take away so that you would see these as equivalent?”



- If price increases, our consumer gets to a lower indifference curve ($u^1 < u^0$). This price change is welfare equivalent to a DECREASE of income from its initial level m^0 to a new level $m' < m^0$ and movement from x^0 to x' . Under old prices, we reach the new utility level u^1 if we consume bundle x' . So to reach new utility level, we have to take away amount $EV = m^0 - m'$, hence Equivalent Variation is positive.
- If price decreases, our consumer gets to a higher indifference curve ($u^1 > u^0$). This price change is welfare equivalent to an INCREASE of income from its initial level m^0 to a new level $m' > m^0$ and movement from x^0 to x' . So to reach new utility level under old prices, we have to GIVE our consumer amount $m' - m^0$, so technically, the amount we have to take away from a consumer is NEGATIVE, i.e. $EV = m^0 - m' < 0$.

Equivalent Variation

Price of x_1 increases: $p_1^0 \rightarrow p_1^1$, p_2 fixed



$$m^0 = p_1^0 x_1^0 + p_2 x_2^0 = p_1^1 x_1^1 + p_2 x_2^1 = e(p_1^0, p_2, u^0) = e(p_1^1, p_2, u^1)$$

$$m' = p_1^0 x_1' + p_2 x_2' = e(p_1^0, p_2, u^1)$$

$$EV = m^0 - m'$$

CV and EV using Expenditure Function

Price change of p_1 : $p_1^0 \rightarrow p_1^1$, other prices p_{-1} fixed

Utility change: $u^0 \rightarrow u^1$

$$\begin{aligned} CV &= e(p_1^1, p_{-1}, u^0) - m = e(p_1^1, p_{-1}, u^0) - e(p_1^1, p_{-1}, u^1) = \\ &= e(p_1^1, p_{-1}, u^0) - e(p_1^0, p_{-1}, u^0) \end{aligned}$$

$$CV = e(p_1^1, p_{-1}, u^0) - e(p_1^0, p_{-1}, u^0) = \int_{p_1^0}^{p_1^1} \frac{\partial e(p, u^0)}{\partial p_1} dp_1 = \int_{p_1^0}^{p_1^1} h_1(p, u^0) dp_1$$

$$\begin{aligned} EV &= m - e(p_1^0, p_{-1}, u^1) = e(p_1^0, p_{-1}, u^0) - e(p_1^0, p_{-1}, u^1) = \\ &= e(p_1^1, p_{-1}, u^1) - e(p_1^0, p_{-1}, u^1) \end{aligned}$$

$$EV = e(p_1^1, p_{-1}, u^1) - e(p_1^0, p_{-1}, u^1) = \int_{p_1^0}^{p_1^1} \frac{\partial e(p, u^1)}{\partial p_1} dp_1 = \int_{p_1^0}^{p_1^1} h_1(p, u^1) dp_1$$

So it's similar to the Hicksian decomposition, but not the same:

- Here we decompose on the NEW indifference curve

Initial income m^0 is:

- minimum spending we need to have to reach utility level u^0 under prices (p_1^0, p_2)
- minimum spending we need to have to reach utility level u^1 under prices (p_1^1, p_2)

Compensated income m' is:

- minimum spending we need to have to reach utility level u^1 under prices (p_1^0, p_2)

This approach is more technical, but don't worry if you do not know much about integrals. I include this slide and following slides to complete an overall picture of the problem covered. Integrals are not essential to understand the topic. You won't be asked to solve any integrals, if you don't want. You could always use the other approach to compute EV and CV.

Note that p_{-1} means all prices but price of good 1.

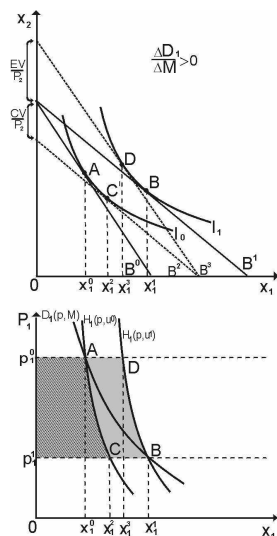
We could use Shepard lemma stating that derivative of expenditure function with respect to p_i is Hicksian demand for good i .

Compensating variation is area under the Hicksian demand between p^0 and p^1 when a consumer wants to reach a utility u^0 .

Equivalent variation is area under the Hicksian demand between p^0 and p^1 when a consumer wants to reach a utility u^1 .

Notice that we could use a more intuitive mathematical approach to draw Hicksian demand when showing CV and EV graphically (so x could be on vertical axis).

CV and EV using Hicksian Demand: Normal Good



Price decrease: $p_1^0 \rightarrow p_1^1$

Original budget line $B^0 \rightarrow$ optimum A

New budget line $B^1 \rightarrow$ optimum B

Points A and B on Walrasian demand $D_1(p, m)$

CV – bring consumer to u_0 under new prices

Point C: optimum when u_0 reached under new prices

A and C on Hicksian demand $h(p, u^0)$

EV – bring consumer to u_1 under old prices

Point D: optimum when u_1 reached under old prices

B and D on Hicksian demand $h(p, u^1)$

$|CV|$ = area below $h(p, u^0)$ inbetween p_1^0 and p_1^1

$|EV|$ = area below $h(p, u^1)$ inbetween p_1^0 and p_1^1

Normal good: $|EV| > |CV|$

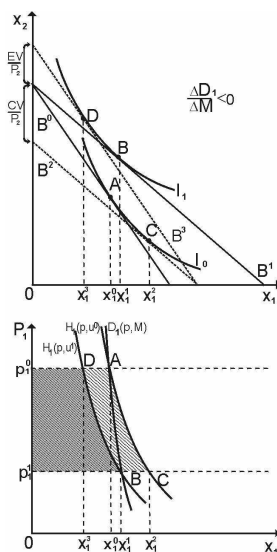
Price decrease $\rightarrow$ EV and CV are negative. So we use absolute values when comparing sizes of CV and EV.

If price decreases, consumer reaches higher level of utility after price decrease, hence $u^1 > u^0$. $|EV| > |CV|$, because EV has u^1 as a base in the integral of Hicksian function.

- And vice versa for price increase.

Areas are drawn to vertical axis (remember that we have inverse demands on our figures!).

CV and EV using Hicksian Demand: Inferior Good



Price decrease: $p_1^0 \rightarrow p_1^1$

Original budget line $B^0 \rightarrow$ optimum A

New budget line $B^1 \rightarrow$ optimum B

Points A and B on Walrasian demand $D_1(p, m)$

CV – bring consumer to u_0 under new prices

Point C: optimum when u_0 reached under new prices

A and C on Hicksian demand $h(p, u^0)$

EV – bring consumer to u_1 under old prices

Point D: optimum when u_1 reached under old prices

B and D on Hicksian demand $h(p, u^1)$

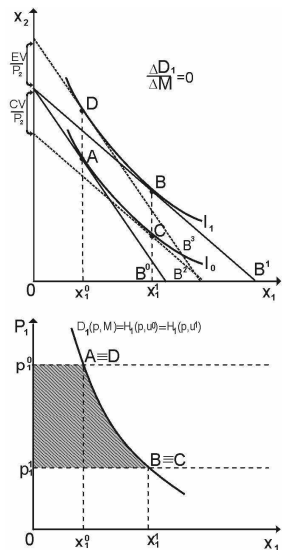
$|CV|$ = area below $h(p, u^0)$ inbetween p_1^0 and p_1^1

$|EV|$ = area below $h(p, u^1)$ inbetween p_1^0 and p_1^1

Inferior good: $|CV| > |EV|$

Analogical (Hicksian demand with new utility lies below old, because income effect is negative for inferior good, so $|CV| < |EV|$)

CV and EV using Hicksian Demand: Quasilinear Preferences



Price decrease: $p_1^0 \rightarrow p_1^1$

Original budget line $B^0 \rightarrow$ optimum A

New budget line $B^1 \rightarrow$ optimum B

Points A and B on Walrasian demand $D_1(p, m)$

CV – bring consumer to u_0 under new prices

Point C: optimum when u_0 reached under new prices

A and C on Hicksian demand $h(p, u^0)$

EV – bring consumer to u_1 under old prices

Point D: optimum when u_1 reached under old prices

B and D on Hicksian demand $h(p, u^1)$

Demands for x_1 identical for points A and D,
and points B and C.

(Zero income effect, quasilinear preferences)

$$h(p, u^1) = h(p, u^0) = D(p, m) \longleftrightarrow |CV| = |EV|$$

$$|CV| = |EV| = \Delta \text{ Consumer surplus}$$

The Hicksian demand for x_1 does not depend on required utility level, so CV and EV are equal with quasilinear preferences.

The Hicksian demands are also identical to the Walrasian demand, so CV and EV are equal to change in consumer surplus.